ARGOFLOW PRO X9

Industrial IoT Edge Controller
Product Guide & Technical Specification

Manufactured by Veltrion Automation Systems
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1. Product Overview

The ArgoFlow Pro X9 is an industrial-grade edge controller designed by Veltrion Automation Systems for real-time monitoring and control of distributed sensor networks in manufacturing, agriculture, and facilities-management environments. The Pro X9 combines a ruggedized enclosure, a multi-protocol connectivity stack, and the ArgoOS 4.3 firmware platform to provide local decision-making at the edge, reducing dependency on constant cloud connectivity.

Unlike general-purpose IoT gateways, the Pro X9 is purpose-built for environments with intermittent connectivity, wide temperature swings, and electrically noisy conditions, such as factory floors, cold-storage facilities, and outdoor agricultural installations. It is the flagship model in the ArgoFlow product family, succeeding the Pro X7 (discontinued December 2025).

1.1 Target Applications

- Predictive maintenance monitoring on industrial machinery.
- Cold-chain temperature and humidity logging for food and pharmaceutical storage.
- Smart irrigation and soil-condition monitoring for precision agriculture.
- Building management system (BMS) integration for HVAC and energy monitoring.
- Remote asset tracking for fleets of stationary equipment.

2. Product Family and Variants

The ArgoFlow Pro X9 is available in three variants, each targeting a different deployment scale and budget.

Variant	I/O Channels	Onboard Storage	Typical Use Case
ArgoFlow Pro X9 Lite	8 channels	4 GB eMMC	Single-machine monitoring, small sites
ArgoFlow Pro X9 (Standard)	16 channels	16 GB eMMC	Standard industrial deployment
ArgoFlow Pro X9 Max	32 channels	64 GB eMMC + microSD slot	Large multi-zone facilities

All variants run the same ArgoOS 4.3 firmware and share the same core connectivity stack described in Section 6. The Max variant additionally supports firmware-level edge-AI inference via the optional Veltrion Inference Module (see Section 16.3).

3. Key Features

3.1 Edge Processing

The Pro X9 runs a quad-core 1.8 GHz ARM Cortex-A72 processor with 4 GB of LPDDR4 RAM (8 GB on the Max variant), enabling local rule evaluation, data aggregation, and buffering without requiring a live connection to a central server.

3.2 Multi-Protocol Support

- Modbus RTU and Modbus TCP for legacy industrial equipment.
- MQTT 5.0 for lightweight cloud telemetry publishing.
- BACnet/IP for building automation integration.
- OPC-UA client and server modes for SCADA interoperability.
- LoRaWAN (via optional radio module) for long-range, low-power sensor networks.

3.3 Local Buffering and Store-and-Forward

During connectivity outages, the Pro X9 buffers up to 30 days of sensor data (Standard variant) in onboard flash storage and automatically forwards the buffered data once connectivity is restored, using a store-and-forward queue with configurable retry intervals.

3.4 Edge Rule Engine

ArgoOS includes a visual rule-builder accessible via the ArgoLink Mobile Companion App (Section 12), allowing operators to define threshold-based alerts and automated relay triggers without writing code.

4. Technical Specifications

Specification	Value
Processor	Quad-core ARM Cortex-A72 @ 1.8 GHz
RAM	4 GB LPDDR4 (8 GB on Max variant)
Onboard Storage	16 GB eMMC (varies by variant; see Section 2)
Operating Temperature	-20°C to +60°C
Storage Temperature	-30°C to +70°C
Humidity Tolerance	5% to 95% RH, non-condensing
Ingress Protection	IP65 (enclosure sealed, connectors capped)
Power Input	12–24V DC, 18W typical, 30W peak
Weight	890 grams (Standard variant)
Firmware	ArgoOS 4.3 (based on a hardened Linux kernel)

5. Physical Dimensions and Mounting

Variant	Dimensions (H x W x D)	Mounting Options
Pro X9 Lite	120mm x 90mm x 40mm	DIN rail, wall bracket
Pro X9 Standard	160mm x 110mm x 45mm	DIN rail, wall bracket, panel mount
Pro X9 Max	200mm x 140mm x 55mm	DIN rail, wall bracket, 19-inch rack adapter

All variants support standard 35mm DIN rail mounting (EN 50022 profile) as the default installation method. Wall-mount brackets and panel-mount kits are available as optional accessories (Section 16).

6. Connectivity and I/O

6.1 Wired Connectivity

- 2x Gigabit Ethernet ports (1x WAN, 1x LAN with built-in switch for daisy-chaining).
- 1x RS-485 port for Modbus RTU devices.
- 1x RS-232 console port for local debugging.
- 1x USB-C port for firmware recovery and diagnostics (not for general peripheral use).

6.2 Wireless Connectivity

- Dual-band Wi-Fi 6 (802.11ax), 2.4 GHz and 5 GHz.
- Bluetooth 5.2 LE for short-range commissioning via the ArgoLink app.
- Optional 4G LTE Cat-6 cellular module (sold separately, see Section 16.1).
- Optional LoRaWAN radio module for long-range sensor mesh (sold separately, see Section 16.2).

6.3 Digital and Analog I/O

I/O Type	Lite	Standard	Max
Digital Inputs	4	8	16
Digital Outputs (Relay)	2	4	8
Analog Inputs (4-20mA / 0-10V)	2	4	8

7. Power Requirements

The Pro X9 accepts a 12–24V DC input via a locking Phoenix-style terminal connector, rated for 18W typical draw and up to 30W peak draw when all I/O channels are active and the optional cellular module is transmitting.

7.1 Power Options

- Direct DC input from a Class 2 power supply (not included; see Section 20 for packaging contents).
- PoE+ (802.3at) via the WAN Ethernet port on the Standard and Max variants, eliminating the need for a separate power cable.
- Optional battery backup module providing up to 4 hours of runtime during a power interruption (sold separately, Section 16.4).

7.2 Power Failure Behavior

On power loss, the Pro X9 performs a graceful shutdown sequence, flushing any pending data writes to onboard storage within 150 milliseconds using onboard supercapacitor reserve power, to prevent data corruption.

8. Environmental and Certification Ratings

Certification	Standard	Scope
Ingress Protection	IP65	Enclosure sealed against dust and water jets
Electromagnetic Compatibility	IEC 61000-6-2 / 6-4	Industrial EMC immunity and emissions
Vibration Resistance	IEC 60068-2-6	Suitable for machinery-adjacent mounting
Hazardous Location	Class I, Division 2 (optional cert.)	Available on Max variant, special order
RoHS Compliance	RoHS 3 (EU 2015/863)	All variants

9. Installation Guide

9.1 Pre-Installation Checklist

- Confirm the mounting location meets the operating temperature range in Section 4.
- Verify available power source matches Section 7 requirements.
- Ensure Ethernet or cellular connectivity is available at the installation site.
- Download the ArgoLink Mobile Companion App (Section 12) on a commissioning device before arriving on-site.

9.2 Step-by-Step Installation

- Step 1 — Mount the unit on a DIN rail, wall bracket, or panel using the method appropriate for the variant (see Section 5).
- Step 2 — Connect the DC power supply or PoE+ Ethernet cable to the WAN port.
- Step 3 — Connect sensor and actuator wiring to the appropriate digital/analog I/O terminals, referring to the terminal block labeling on the unit's underside.
- Step 4 — Power on the unit; the status LED will blink amber during boot (approximately 45 seconds) and turn solid green once ArgoOS has fully initialized.
- Step 5 — Open the ArgoLink app, select "Add Device", and pair via Bluetooth using the QR code printed on the unit's side label.
- Step 6 — Follow the in-app commissioning wizard to assign the unit a site name, configure network settings, and map connected sensors.

9.3 Post-Installation Verification

After commissioning, confirm the status LED is solid green and that the ArgoLink app's "Device Health" screen shows all connected I/O channels as "Active" with no fault codes. Refer to the companion Support Procedures document for troubleshooting any fault codes encountered during this step.

10. Initial Configuration

10.1 Network Configuration

By default, the Pro X9 attempts DHCP on the WAN Ethernet port. Static IP configuration is available through the ArgoLink app under Settings > Network, or via the local web console at https://192.168.1.1 (default) when connected directly to the LAN port.

10.2 Default Credentials

The unit ships with a default administrator username of "admin" and a unique default password printed on the label affixed to the underside of the unit. Veltrion strongly recommends changing this password during initial commissioning, as described in Section 10.3.

10.3 Changing the Administrator Password

- Navigate to Settings > Security > Administrator Account in the ArgoLink app.
- Enter the current (default) password.
- Enter a new password meeting the minimum complexity requirement of 12 characters, including at least one number and one symbol.
- Confirm the change; the unit will apply the new credential immediately without requiring a reboot.

11. ArgoOS Firmware Overview

ArgoOS 4.3 is Veltrion's hardened Linux-based firmware platform purpose-built for the ArgoFlow product line. It provides the rule engine, protocol stack, local web console, and over-the-air (OTA) update mechanism used across all Pro X9 variants.

11.1 Firmware Update Channels

Channel	Update Frequency	Recommended For
Stable	Quarterly	Production deployments
Beta	Monthly	Pre-production testing
LTS (Long-Term Support)	Security patches only, 3-year support window	Regulated or critical environments

11.2 OTA Update Process

Firmware updates are delivered over-the-air by default when the unit has internet connectivity, with update windows configurable to avoid disrupting production hours. Updates are staged and verified via checksum before being applied, and the unit automatically rolls back to the previous firmware version if the new version fails to boot within 3 attempts.

12. The ArgoLink Mobile Companion App

ArgoLink is Veltrion's companion application, available for iOS and Android, used for commissioning, configuration, monitoring, and basic troubleshooting of ArgoFlow devices.

12.1 Key Screens

- Device Health — real-time status of connectivity, I/O channels, and buffered data queue.
- Rule Builder — visual, no-code interface for defining threshold alerts and relay automation.
- Data Explorer — historical charting of sensor readings over configurable time ranges.
- Settings — network, security, and firmware update configuration.

12.2 Multi-Site Management

For deployments spanning multiple sites, ArgoLink supports a "Fleet View" mode, allowing a single administrator account to monitor and manage up to 200 Pro X9 units across different physical locations from one dashboard.

13. Sensor Compatibility Matrix

Sensor Type	Interface	Compatibility
Temperature/Humidity Probes	4-20mA, RS-485 Modbus	Native, no adapter needed
Vibration Accelerometers	4-20mA	Native
Pressure Transducers	0-10V, 4-20mA	Native
Soil Moisture Sensors	RS-485 Modbus, LoRaWAN	Native / requires LoRaWAN module
Legacy 0-5V Analog Devices	0-5V	Requires Veltrion Signal Adapter (sold separately)

14. Integration with Third-Party Platforms

The Pro X9 is designed to integrate with common industrial and cloud platforms without requiring custom middleware in most cases.

- AWS IoT Core — native MQTT publishing with certificate-based authentication.
- Azure IoT Hub — supported via MQTT with SAS token authentication.
- Generic SCADA systems — via OPC-UA server mode or Modbus TCP gateway mode.
- Building Management Systems — via BACnet/IP object mapping configured through ArgoLink.
- Custom REST APIs — via the ArgoOS Webhook Forwarder, configurable per data point.

15. Safety Information

15.1 General Safety Warnings

- Disconnect power before performing any wiring changes to the I/O terminals.
- Do not install the Pro X9 in locations exceeding the environmental ratings specified in Section 8 unless using the certified Class I, Division 2 variant.
- The relay outputs are rated for a maximum switching load of 5A at 250V AC; exceeding this rating may cause permanent damage or fire risk.
- Do not open the enclosure in the field; doing so voids the IP65 rating and the product warranty.

15.2 Electrical Safety

All wiring must be performed by a qualified electrician or technician familiar with the relevant local electrical codes. Veltrion Automation Systems is not liable for damage or injury resulting from improper installation.

16. Accessories and Optional Modules

16.1 Cellular Connectivity Module

The ArgoCell LTE Module adds 4G LTE Cat-6 connectivity for sites without reliable wired or Wi-Fi infrastructure. Installs via the internal expansion slot; requires a compatible nano-SIM card (sold separately by local carriers).

16.2 LoRaWAN Radio Module

The ArgoMesh LoRaWAN Module enables connectivity with long-range, low-power sensors up to 10 km away in line-of-sight conditions, ideal for large agricultural or campus deployments.

16.3 Veltrion Inference Module

Available exclusively for the Pro X9 Max variant, this module adds a dedicated neural processing unit (NPU) capable of running lightweight anomaly-detection models directly on the device, reducing the need to stream raw data to the cloud for analysis.

16.4 Battery Backup Module

Provides up to 4 hours of runtime during a power interruption, automatically engaging within 10 milliseconds of a detected power loss.

16.5 Mounting Accessories

Accessory	Part Number	Compatible Variants
Wall Mount Bracket Kit	VAS-ACC-WM1	All
19-inch Rack Adapter	VAS-ACC-RK2	Max only
Outdoor Enclosure Hood	VAS-ACC-OH3	All

17. Use Case Scenarios

17.1 Cold Storage Monitoring

A food distribution warehouse deploys 6 Pro X9 Standard units across cold-storage rooms, each monitoring temperature and humidity via Modbus-connected probes. Rule Builder alerts notify facility managers via SMS and email if any room deviates from its target range for more than 10 minutes, with all readings logged locally during any connectivity outage and automatically synced once restored.

17.2 Precision Agriculture

A vineyard operation uses Pro X9 Lite units fitted with the ArgoMesh LoRaWAN module to collect soil moisture readings from sensors spread across a 40-hectare property, enabling targeted irrigation scheduling that reduced water usage by a reported 18% in the first season.

17.3 Predictive Maintenance

A manufacturing plant installs Pro X9 Max units with the Veltrion Inference Module near critical rotating machinery, using onboard vibration-pattern analysis to flag early signs of bearing wear before a failure occurs, reducing unplanned downtime.

18. Comparison with Previous Generation (Pro X7)

Specification	Pro X7 (Discontinued)	Pro X9 (Current)
Processor	Dual-core 1.2 GHz Cortex-A53	Quad-core 1.8 GHz Cortex-A72
Max I/O Channels	24	32
Wi-Fi Standard	Wi-Fi 5 (802.11ac)	Wi-Fi 6 (802.11ax)
Local Buffer Capacity	10 days	30 days
Edge AI Support	Not available	Available (Max variant, optional module)
Ingress Protection	IP54	IP65

19. Warranty Terms

Veltrion Automation Systems provides a standard 3-year limited hardware warranty on all ArgoFlow Pro X9 units from the date of original purchase, covering manufacturing defects under normal use conditions.

19.1 What Is Covered

- Manufacturing defects in materials or workmanship.
- Premature hardware failure under rated operating conditions.
- Firmware-related faults not resolved by a standard OTA update.

19.2 What Is Not Covered

- Damage from installation outside the rated environmental specifications in Section 8.
- Damage from unauthorized enclosure opening or modification.
- Damage from incorrect wiring, over-voltage, or exceeding relay switching ratings.
- Consumable accessories such as SIM cards or third-party sensors.

19.3 Extended Warranty

An optional 2-year warranty extension (VAS-WAR-EXT2) is available for purchase within 90 days of the original purchase date, extending total coverage to 5 years.

20. Packaging Contents

Item	Included
ArgoFlow Pro X9 Unit	Yes
DIN Rail Mounting Clip	Yes
Terminal Block Connector Set	Yes
Quick Start Guide (printed)	Yes
DC Power Supply	No — sold separately
Ethernet Cable	No — sold separately
SIM Card (for cellular module)	No — sold separately, carrier-provided

21. Frequently Asked Questions

Q1. Can I upgrade a Pro X9 Standard to a Max variant?

No. The variants differ in hardware (processor, I/O count, storage) and cannot be upgraded in the field. Customers should select the appropriate variant at time of purchase based on projected I/O requirements.

Q2. Does the Pro X9 work without an internet connection?

Yes. The onboard rule engine and local buffering (Section 3.3, 3.4) allow the unit to continue monitoring and triggering relay actions locally even without connectivity, buffering data until connectivity is restored.

Q3. What happens if I lose the default admin password and never changed it?

Contact Veltrion Support with proof of purchase and the unit's serial number to initiate a factory reset procedure, detailed in the companion Support Procedures document.

Q4. Can I mix sensor types across I/O channels?

Yes. Each digital and analog I/O channel can be independently configured for a different sensor type through the ArgoLink Rule Builder; there is no requirement for uniform sensor types across channels.

22. Glossary of Terms

Term	Meaning
ArgoOS	The firmware platform running on all ArgoFlow Pro X9 units.
ArgoLink	The mobile companion app used for commissioning and monitoring.
Store-and-Forward	Local data buffering during connectivity loss, with automatic sync on reconnection.
NPU	Neural Processing Unit; dedicated hardware for on-device AI inference.
OTA	Over-the-Air; remote firmware update delivery method.
DIN Rail	A standardized metal rail used for mounting industrial control equipment.

All product names, specifications, and company details in this document are fictional and created for software testing and training purposes.